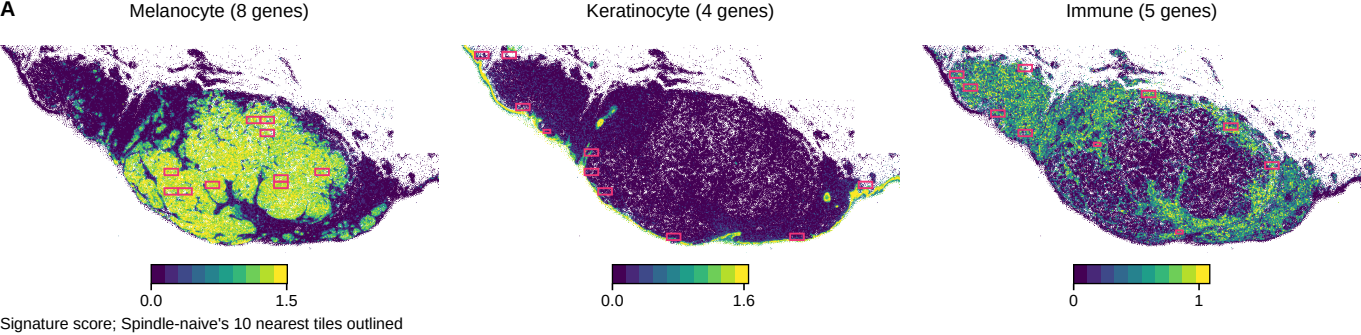
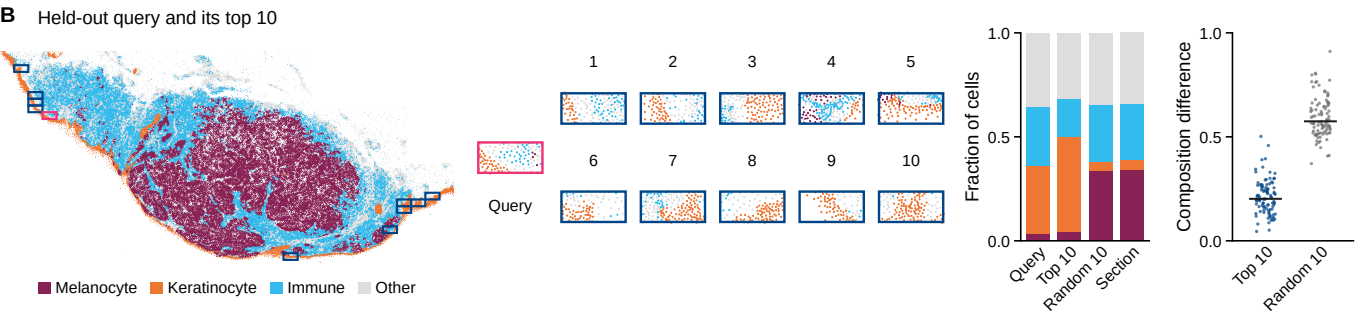
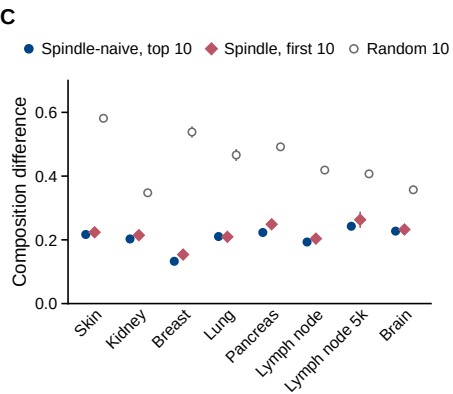
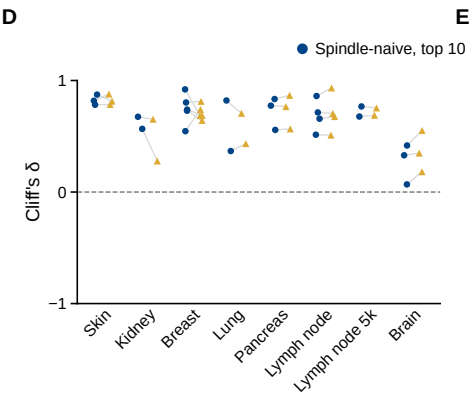
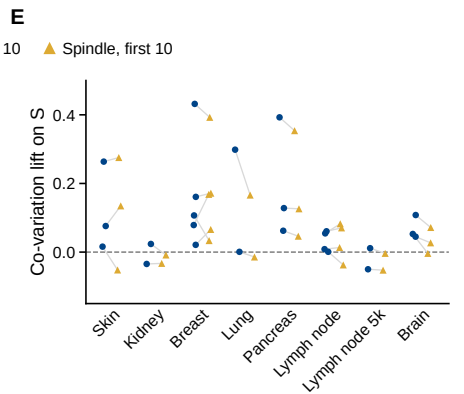

A
Melanocyte (8 genes)
Keratinocyte (4 genes)
Immune (5 genes)
0.0
1.5
0.0
1.6
0
1
Signature score; Spindle-naive's 10 nearest tiles outlined


B
Held-out query and its top 10
1
2
3
4
5
6
7
8
9
10
Query
Melanocyte
Keratinocyte
Immune
Other
Fraction of cells
1.0
0.5
0.0
Query
Top 10
Random 10
Section
Composition difference
1.0
0.5
0.0
Top 10
Random 10


C
Spindle-naive, top 10
Spindle, first 10
Random 10
Composition difference
0.6
0.4
0.2
0.0
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain


D
Cliff's δ
1
0
−1
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain


E
Spindle-naive, top 10
Spindle, first 10
Co-variation lift on S
0.4
0.2
0.0
Skin
Kidney
Breast
Lung
Pancreas
Lymph node
Lymph node 5k
Brain